LeepTalent Level 3 End-point Assessment

Learner Guide

Employment by Industry — what to do, why, and where you are in the process | ★★★★★

This guide shows one sound way to complete the scenario, the reasoning behind each step, and which stage of the data analysis process each step belongs to. This is the hardest scenario in the series — parts of Scenario 2 (correlation, forecasting) have no single right answer, so the guide shows a defensible method and explains the judgement involved. Work the task yourself first, then check your approach here. Screenshots are illustrative mock-ups; Excel/Power Query is shown.

> **The data analysis process — follow it in order:**
>
> **Identify → Collect → Clean/Prepare → Analyse → Visualise → Interpret/Report**
>
> Each step is tagged with its stage. On a messy data set like this, most of the marks sit in the Clean/Prepare stage — get that right and the analysis follows.

Scenario 1 — Data Gathering and Validation

Step 1 — Identify what you need

STAGE 1 · IDENTIFY

What you do: Confirm the requirement first: two sources (`men.csv` , `women.xlsx`) plus the Annex A quarter-to-date table; the goal is one combined, cleaned set with a Gender column and proper dates. Note the known issues up front — multi-row headers, ".." placeholders, comma-formatted numbers, sector data only from 1997.

> **Why start here?** This data is messy. Knowing the structural problems before you start means you plan the reshape rather than discovering issues halfway through and redoing work.

Step 2 — Bring each source in as a connection

STAGE 2 · COLLECT

What you do: Import each source as a **connection only**:

1. `Data` → `Get Data` → `From File` → the CSV or the Excel workbook.
2. At the load prompt, choose `Only Create Connection` .

Import Data ×

Select how you want to view this data in your workbook.

- ○ Table
- ○ PivotTable Report
- ○ PivotChart
- ● **Only Create Connection**

↑ *Keeps each source as a query — clean it before loading.*

OK | Cancel

*Figure 1 — Choose **Only Create Connection** so each source stays as a query to be reshaped.*

> **Why connection only?** These files need real reshaping before they're usable. Loading them straight to a sheet would dump the messy headers into the workbook; a connection-only query lets you clean inside Power Query first.

Step 3 — Reshape the messy headers

STAGE 3 · CLEAN / PREPARE

What you do: Double-click a query to open it in Power Query, then fix the structure:

1. `Remove Top Rows` to strip the title and grouping rows above the data.
2. `Use First Row as Headers` to promote the real column names.
3. Replace the **..** placeholder values with `null`.
4. Set data types: the quarter column as text, employment columns as whole numbers (this also strips the thousands separators).

Power Query Editor

Remove Top Rows | **Use First Row as Headers**

Column1	Column2	Column3	Column4
(null)	Standard Industrial Class...	(null)	(null)
(null)	All in employment	Public sector	Private sector
(null)	(null)	(null)	(null)
Jan-Mar 1995	13,941	..	..
Apr-Jun 1995	14,066	..	..

⚠ The red rows are header/grouping junk above the data — remove them, then promote the real header row.

".." marks periods with no sector breakdown (pre-1997) — replace with null so figures stay numeric.

Figure 2 — Removing the header/grouping rows, then promoting the real header. The ".." values become null.

> **Why reshape before anything else?** Until the header is correct and the numbers are numeric, no formula, chart or merge will work. Treating ".." as text or as zero is a classic error — as text it breaks calculations; as zero it understates early figures. Replacing it with null keeps the column numeric while correctly recording "no data".

Step 4 — Append the two series and add Gender

STAGE 3 · CLEAN / PREPARE

What you do: Add a **Gender** column to each query ("Male" / "Female"), then `Append Queries` to stack them into one table. Remove any duplicate rows and check the industry names match across both.

> **Why append, not merge here?** The two files have the *same* columns for different people, so they stack (append) on top of each other — unlike the earlier packs where different columns were merged side by side. The Gender column is what lets you separate them again during analysis.

Step 5 — Standardise the date

STAGE 3 · CLEAN / PREPARE

What you do: Convert each quarter label to the **last day of the quarter** using the Annex A lookup. In Excel:

```
=DATE(RIGHT(A2,4), VLOOKUP(LEFT(A2,7),QtrTable,2,0),
VLOOKUP(LEFT(A2,7),QtrTable,3,0))
```

C2 fx `=DATE(RIGHT(A2,4), VLOOKUP(LEFT(A2,7),QtrTable,2,0), VLOOKUP(LEFT(A2,7),QtrTable,3,0))`

Quarter	All in employment	Quarter End Date
Jan-Mar 1995	13,941	31/03/1995
Apr-Jun 2002	15,402	30/06/2002
Oct-Dec 2018	16,701	31/12/2018

A lookup maps each quarter to its end-of-quarter day; DATE() builds a real date so it sorts and charts correctly.
QtrTable is the Annex A reference loaded as a small lookup range.

Figure 3 — Mapping each quarter to its end-of-quarter date so the series sorts and charts correctly.

> **Why a real date, not the text label?** "Jan-Mar 1995" can't be sorted, filtered by range or charted on a time axis. Converting to 31/03/1995 turns it into a true date the tool understands. Use the **last** day of the quarter so each point sits at the end of the period it represents. Save as `Employment_Cleaned.xlsx`.

Scenario 2 — Data Analysis

Note on this scenario: several tasks (correlation, forecasting) are open-ended. There isn't one correct number — what's assessed is whether you chose a sensible method, applied it correctly, and explained your reasoning. The figures below are indicative of what the data shows.

Step 6 — Public and private sector trends (1997–2023)

STAGE 4 · ANALYSE

What you do: Filter to 1997 onward, then chart public and private employment over time and summarise with average/max/min.

Indicative result (men): private-sector employment rises from about **11,851** (Jan-Mar 1997) to **14,310** (Apr-Jun 2023); public sector moves far less, from about **2,327** to **2,844**. The growth is overwhelmingly in the private sector.

Step 7 — Gender comparison and sector focus

STAGE 4 · ANALYSE

What you do: Compare male and female totals over time (side-by-side bars or two lines), and extract the **Mining, energy & water supply** series for a male-vs-female sector comparison.

Indicative result: total employment rises for both, but women's grows faster — men +3,303 (13,941 → 17,244) versus women +4,047 (11,576 → 15,623) across 1995–2023 — so the gap narrows over time (convergence).

Step 8 — Correlation analysis

STAGE 4 · ANALYSE

What you do: Test whether sectors move together — e.g. private vs public — with `=CORREL(range1,range2)` and a scatter plot, or a correlation matrix across several sectors.

> **Why correlation, and what it does and doesn't tell you:** a correlation coefficient quantifies whether two series move together (near +1), move oppositely (near -1) or are unrelated (near 0). State clearly that correlation is not causation — two sectors rising together may both be driven by overall economic growth rather than by each other.

Step 9 — Visualise trends and forecast to 2028

STAGE 5 · VISUALISE

What you do: Plot the cleaned series as a line chart, then extend it five years using a forecast method — Excel's **Forecast Sheet**, a `FORECAST.LINEAR` / `TREND` formula, or a fitted trend line.

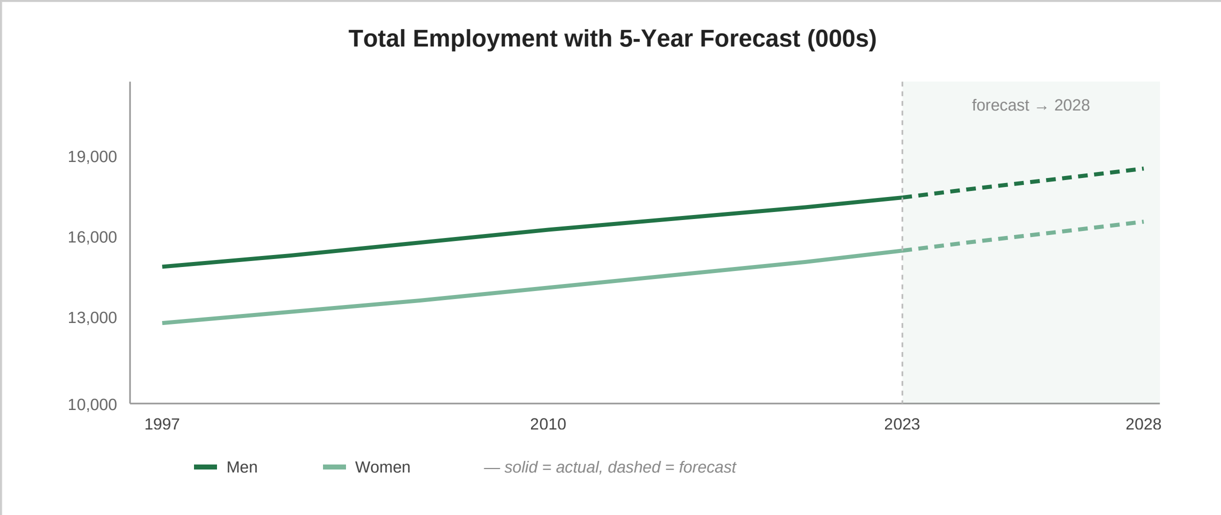

Figure 4 — Total employment with a five-year forecast (dashed) to 2028.

> **Why state your forecast assumptions?** A linear forecast assumes the recent trend simply continues — it can't predict shocks (recessions, policy changes). Showing the forecast as a dashed extension and noting the assumption is what separates a credible projection from a guess. This is exactly the judgement assessors reward at this level.

Step 10 — Interpret and report

STAGE 6 · INTERPRET / REPORT

What you do: Summarise the findings for senior management beneath your charts, then save the workbook with its tables, charts and forecast intact.

> "Over 1995–2023, total employment grew for both sexes, with women's rising faster (+4,047 vs +3,303), narrowing the gender gap. Growth was almost entirely in the private sector, while public-sector employment stayed broadly flat. Sector series move largely together, consistent with overall economic growth rather than substitution between sectors. On a continuation of the recent trend, total employment is projected to keep rising modestly to 2028 — though this assumes no major economic shock."

Marker's checklist — process followed in order

- **Identify:** sources, goal and the known data problems confirmed first.
- **Collect:** data brought in as **connection only** (not opened); double-clicked to open.
- **Clean/Prepare:** header rows removed and real header promoted; ".." handled as null; numbers parsed; series appended with a Gender column; quarters converted to end-of-quarter dates.
- **Analyse:** public/private trends; gender and sector comparisons; correlation tested.
- **Visualise:** line/bar charts; a clearly-marked five-year forecast.
- **Interpret/Report:** findings summarised with stated forecast assumptions; saved as .xlsx.